

Worked Solutions

Data Analytics

Set A: Ensemble | CSE Semester 3

Companion to da-ensemble.pdf. Question numbers match that paper. Equivalent correct methods are acceptable. These are independent worked solutions, not an official university marking scheme.

Questions 1–5: MCQ answer key, 1 mark each (5 marks total). Questions 6 onward: theory solutions (25 marks total).

1. Null hypothesis

Correct option: (A). The null hypothesis specifies the population parameter: $H_0 : \mu = 0.95$.

2. Standard error

Correct option: (C). $SE = \sigma / \sqrt{n} = 0.02 / \sqrt{36} = 0.02/6$.

3. Normal-test statistic

Correct option: (B). $z = (0.947 - 0.95) / (0.02/6) = -0.90$.

4. Equal-width bins

Correct option: (D). Bin width is $(480 - 180) / 3 = 100$.

5. Equal-frequency bins

Correct option: (B). Each bin contains $12/3 = 4$ observations.

6. Descriptive statistics, preparation and ANOVA

(i) $N = 20$, $\sum x_i = 1350$, $\bar{x} = 67.5$, and the median is 63. Using the population denominator,

$$\sigma^2 = \frac{\sum (x_i - 67.5)^2}{20} = 238.35, \quad \sigma = 15.4386.$$

Pearson's second coefficient is $3(67.5 - 63) / 15.4386 = 0.8744$, indicating positive skewness.

(ii) Combine quantity and unit price to create revenue: $\text{revenue} = \text{quantity} \times \text{unit price}$. This gives a useful derived variable from two existing columns.

(iii) Test $H_0 : \mu_A = \mu_B = \mu_C$ against at least one unequal mean. The group means are 3, 5 and 7; the grand mean is 5.

$$SS_B = 4[(3 - 5)^2 + (5 - 5)^2 + (7 - 5)^2] = 32, \quad SS_W = 4 + 2 + 4 = 10.$$

Source	SS	df	MS	F
Between	32	2	16	14.4
Within	10	9	1.1111	
Total	42	11		

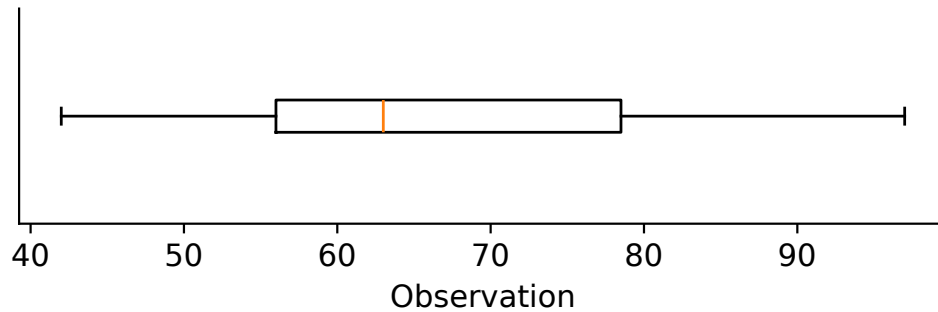
Since $14.4 > 4.256$, reject H_0 at 5%. Mean flavour ratings differ. The test alone does not identify which pairs differ.

7. Box plot and Spearman correlation

(i) Sorted observations:

42, 47, 50, 50, 54, 58, 60, 61, 62, 63, 63, 68, 75, 78, 78, 79, 82, 91, 92, 97.

The five-number summary is $[42, 56, 63, 78.5, 97]$. Here $Q_1 = (54 + 58)/2 = 56$, $Q_3 = (78 + 79)/2 = 78.5$, and $IQR = 22.5$. The fences are 22.25 and 112.25, so there are no outliers.



(ii) Subsets support analysis of relevant cohorts, reduce computational load, and allow separate training and validation sets. Select them appropriately to avoid introducing bias.

(iii) The first two ranks agree. The other ten items have rank differences ± 1 , so $\sum d_i^2 = 10$.

$$r_s = 1 - \frac{6(10)}{12(12^2 - 1)} = \boxed{0.9650}.$$

The interviewers have strong positive rank agreement.

8. Left-tailed normal test

Under the supplied model, $H_0 : \mu = 0.95$ and $H_1 : \mu < 0.95$.

$$z = \frac{0.93 - 0.95}{0.04/\sqrt{50}} = \boxed{-3.5355}.$$

Since $-3.5355 < -1.645$, reject H_0 . The evidence supports a mean accuracy below 95% under this model ($p \approx 0.000203$). The supplied standard deviation is treated as that of accuracy measurements, not inferred as a Bernoulli standard deviation from individual positive/negative results.

9. Vaccination and disease: chi-square test

H_0 : vaccination status and disease status are independent. H_1 : they are associated. Compute each expected count as row total times column total divided by 2000.

Group	Attacked: observed / expected	Not attacked: observed / expected
Vaccinated	31 / 54	469 / 446
Not vaccinated	185 / 162	1315 / 1338

$$\chi^2 = \frac{23^2}{54} + \frac{23^2}{446} + \frac{23^2}{162} + \frac{23^2}{1338} = \boxed{14.6432}.$$

There is one degree of freedom. Since $14.6432 > 3.841$, reject independence. The observed attack rates are $31/500 = 6.2\%$ and $185/1500 = 12.33\%$. Vaccination is associated with a lower attack rate in this sample; the table alone does not establish causation.

10. Two-factor ANOVA with replication

There are $a = 3$ varieties, $b = 4$ fertilizers and $r = 2$ replicates per cell. The grand mean is 8.6667. Variety means are (8.5, 7.25, 10.25) and fertilizer means are (8.1667, 9.5, 6.1667, 10.8333). Use

$$SS_A = br \sum_i (\bar{y}_{i..} - \bar{y})^2, \quad SS_B = ar \sum_j (\bar{y}_{.j.} - \bar{y})^2,$$

$$SS_{AB} = r \sum_{i,j} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y})^2,$$

$$SS_E = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2.$$

Source	SS	df	MS	F
Variety	36.3333	2	18.1667	36.3333
Fertilizer	71.3333	3	23.7778	47.5556
Interaction	3.6667	6	0.6111	1.2222
Error	6	12	0.5	
Total	117.3333	23		

Reject equal variety means because $36.3333 > 3.885$. Reject equal fertilizer means because $47.5556 > 3.490$. Both factors have significant effects at 5% under the supplied assumptions. The interaction is retained in the model; its statistic is shown for completeness. The residual error mean square, rather than the interaction mean square, is the denominator for these fixed-factor tests.

11. Cholesterol binning

Range = $260 - 145 = 115$, so width = $115/3 = 38.3333$.

Method	Bin 1	Bin 2	Bin 3
Equal width	[145,183.333): 7	[183.333,221.667): 3	[221.667,260]: 2
Equal frequency	145,155,160,162	165,172,180,190	200,220,240,260

For equal frequency, use boundaries 145, 163.5, 195 and 260. Equal width better displays the declining concentration across these data and the upper tail. Equal frequency gives balanced groups but requires density heights to make its unequal-width histogram meaningful.

